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 Fireblocks Help Center

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Fireblocks upgradeable smart contracts

Note

Tokenization is a premium feature that requires an additional purchase. If you don't have access to this feature, contact your [Customer Success Manager](#) to discuss having it enabled in your Fireblocks workspace.

Overview

Fireblocks has written a set of reference smart contracts that implement known ERC standards and provide common token capabilities. They have all been audited by OpenZeppelin.

- [ERC-20F Fungible token](#)
- [ERC-721F Non-fungible token](#)
- [ERC-1155F Semi-fungible token](#)
- [Allowlist](#)
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Upgradeable ERC-20F

A fungible token that can be used as a unit of account, for issuing stablecoins, or for fundraising.

Standards: [ERC-20](#)

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UUPS proxy pattern

1. A *Logic contract* is deployed by Fireblocks that encapsulates the expected token behavior, such as mint, burn, transfer, approve, and so on. These behaviors are also known as the functions in a smart contract.
2. A *Proxy contract* is deployed by you, and it points to the Logic contract. This Proxy contract is the first contract to intercept calls from you or the end user. It is also responsible for storing state, also known as on-chain data. This data is considered the variables in a smart contract.
3. To facilitate upgradeability, the Proxy contract can point to a new Logic contract that contains new behavior. You can redirect your Proxy contract by calling its `upgradeTo()` function from the Fireblocks Console.

Note

Although there are two contracts at work, we simplify this in the Fireblocks platform by allowing customers to operate their contract as if it were one. In the image above, the Upgradeable ERC-20F contract, when you deploy it, is actually a Proxy that references the Logic contract deployed previously by Fireblocks.

Upgrading your contract

If Fireblocks upgrades its reference contracts, we would provide an audited Logic contract with the new behavior that customers could opt into by calling the `upgradeTo()` function. This would point their Proxy contract to the new Logic contract.

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Destination

0xb00CD67d0F0c95A2deB3D1E49DC407f0Eb424E0E

New implementation

Enter an address

Tentative fee 

1.013

GWEI

Slow **Medium** Fast Custom

Note

Submit 

Fireblocks contracts vs standard protocols

The Fireblocks ERC-20F, ERC-721F, and ERC-1155F contracts are enhanced implementations of the standard token protocols on which they are based. While the ERC-20 standard has been the foundation for Ethereum-based tokens, the ERC-20F contract (in addition to the ERC-721F and ERC-1155F contracts) extends its functionality with additional features. This implementation is designed to address the needs of enterprise-level applications in the blockchain ecosystem. These enhancements aim to improve security, flexibility, and operational efficiency in token management.

The following comparison table examines how the Fireblocks smart contracts differ from the traditional token standards, highlighting their potential benefits for businesses operating in the blockchain space.

Feature	Feature Description	ERC-20, ERC-721, ERC-1155	Fireblocks ERC-20F, ERC-721F, ERC-1155F
TotalSupply	Returns the total number of tokens in circulation	Yes	Yes

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Transfer	send tokens to another address	Yes	Yes
TransferFrom	Enables the transfer of tokens from one address to another, typically used with approved spending	Yes	Yes
Approve	Grants permission for a spender to withdraw tokens from the owner's account	Yes	Yes
Allowance	Returns the number of tokens a spender is allowed to withdraw from the owner	Yes	Yes
Mint/Burn	Enable the creation and destruction of tokens	Yes	Yes
Access Registry	Allows or restricts addresses from interacting with the tokens through either an Allow List or Deny List smart contract	No	Yes
Pause/Unpause	Allows authorized role the ability to temporarily halt, and later reinstate all token transfers and operations	No	Yes
Upgrade	Enables authorized role to replace the existing contract logic with a new implementation	No	Yes
Role-based Access Controls	Different functions within the contract can be restricted to specific roles	No	Yes

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Batch Transactions	Allows multiple token transfers or operations to be executed in a single transaction	No	Yes
Fee on Transactions	Enables automatic deduction of a small percentage of each token transfer as a fee	No	No
Token Recovery	Allows for tokens to be retrieved from denylisted addresses	No	Yes
Token Salvage	A safety feature that enables the retrieval of tokens accidentally sent to the contract address	No	Yes

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Yes

No

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